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CIS 344

Final Project Overview

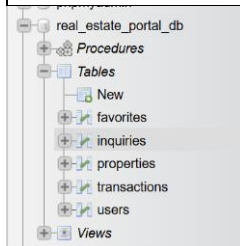
Project Purpose:

The purpose of this project was to develop a web-based Real Estate Agency Portal using PHP and MySQL. The system allows different users such as agents, buyers, and renters to interact with property listings through a role-based platform. Agents can add property listings, while buyers and renters can browse properties and submit inquiries. The project demonstrates database integration, authentication, session management.

Database Design:

The project uses a MySQL database named `real_estate_portal_db`. Multiple related tables were created to manage users, properties, inquiries, favorites, and transactions. Primary keys and foreign keys were used to maintain relationships between tables. The following are the tables I included.

Table Name	Purpose
Users	Stores login and role information
Properties	Stores property listing
Inquiries	Store buyer/renter inquiries
Favorites	Stores saved properties
Transactions	Stores property transactions



PHP Functionality:

PHP was used to connect the application to the MySQL database and handle all backend functionality. Prepared statements were used for database security. The

application supports viewing properties, displaying property details, user authentication, inquiry submission, and property creation by agents.

Login and Session Handling:

The login system validates usernames and passwords using PHP sessions. After successful login, session variables store user information and maintain authentication across pages. Logout destroys the session and prevents unauthorized access.

Role Based Access Control:

The system uses role-based access control to display different features depending on the user type. Agents can add properties, while buyers and renters can browse properties and submit inquiries. Unauthorized users are redirected to the login page when attempting to access protected pages.

Challenges and Solutions:

Several challenges occurred during development, including database field mismatches, session handling issues, and PHP syntax errors. These problems were solved by checking database column names, correcting SQL queries, and debugging PHP code carefully. Testing with phpMyAdmin helped verify that records were correctly inserted into the database.

Screenshots of a Working Platform:

Real Estate Agency Portal

[Home](#) [Properties](#) [Login](#) [Register](#)

Login

Username

Password

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Real Estate Agency Portal

[Home](#) [Properties](#) [Login](#) [Register](#)

Register

Username

peter

Contact Info

4444444444

Password

....

User Type

Buyer



Register

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Real Estate Agency Portal

[Home](#) [Properties](#) [Login](#) [Register](#)

Register

Registration successful. You may now log in.

Username

Contact Info

Password

User Type

Select role



Register

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Property Listings

hello Type: house Address: 123 City: ny Price: \$5000000.00 Status: available Agent: leh View Details
test property Type: house Address: 100 main street City: boston Price: \$50000.00 Status: available Agent: agent1 View Details
Beach Condo Type: Condo Address: 78 Ocean Dr City: Miami Price: \$350000.00 Status: sold Agent: agent1 View Details

Real Estate Agency Portal

[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Dashboard

Welcome: peter

Role: buyer

Common Actions

[Browse Properties](#)